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# Disfluencies reduce the effect of uh ... word surprisal during narrative comprehension



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## ABSTRACT

Disfluencies in speech frequently occur before the production of longer and more complex speech content. Listeners are thought to use the distribution of disfluencies in the comprehension of speech to inform their predictions. Here, we investigated whether the presence of disfluencies in speech affects word processing also in naturalistic listening conditions. Participants ( $n = 36$ ) listened to the spoken recall of the events of a television series while undergoing fMRI. We modelled word processing effort using parametric modulations for word length, frequency, entropy, as well as surprisal and presence/absence of a disfluency. To investigate the effects of disfluencies on word processing, we tested the interaction between disfluency and frequency, and disfluency and surprisal. Words preceded by a disfluency were associated with increased activity in the left and right superior temporal gyrus (STG). Lower word frequency was associated with an increase in activity in the left mid STG. Increased word surprisal elicited a similar distribution of activity, with bilateral superior temporal activation. The effect of surprisal was reduced after a disfluency in a cluster in the left posterior temporal lobe, while the effect of frequency increased following disfluencies in the left superior temporal gyrus and the left inferior frontal cortex. Therefore, the presence of a disfluency affects the response to upcoming input, suggesting that it prepares the listener for higher complexity in the upcoming speech, by potentially allocating increased attention resources that facilitate integration in context.

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## 1. Introduction

The last decade has seen increased attention towards naturalistic language processing in neuroimaging studies of language (Hasson, Egidi, Marelli, & Willems, 2018). Studies have focused both on ecologically valid visual contexts, taking advantage of virtual reality to simulate real-world interacting contexts while maintaining high experimental control (Heyselaar, Peeters, & Hagoort, 2021; Huizeling, Peeters, & Hagoort, 2021, 2023; Peeters, 2019; Tromp, Peeters, Meyer, & Hagoort, 2018), and on rich linguistic contexts by investigating linguistic processing in narrative comprehension (e.g. audiobook listening or book reading) (e.g. Armeni, Willems, van den Bosch, & Schoffelen, 2019; Coopmans, Hoop, Tezcan, Hagoort, & Martin, 2025; Giglio, Ostarek, Sharoh, & Hagoort, 2024; Heilbron, Armeni, Schoffelen, Hagoort, & de Lange, 2022; Wehbe et al., 2014). Both these approaches have the advantage of studying language processing while simulating as much as possible the complex visual and linguistic conditions in which language is exchanged in real life. One critical aspect of natural language that has not received as much attention in the neuroimaging literature is the presence of disfluencies. Disfluencies, which can be manifested as silent pauses, fillers (“uhm” or “uh”), repetitions and repairs, are found to occur on average six times every 100 words (Bortfeld, Leon, Bloom, Schober, & Brennan, 2001). They are used for interpersonal coordination in conversation, such as for turn-taking management and as collaborative tools, and are associated with heavier speech planning demands. Disfluencies are not just relevant from the speaker's perspective, but they are also thought to influence listeners. In this fMRI study of comprehension of spontaneous speech, we investigate how disfluencies affect the neural response to naturalistic language.

In general, disfluencies signal difficulty with conceptual and linguistic planning. They are found more often at the beginning of sentences and phrases (Clark & Fox Tree, 2002) and often occur with message-level, grammatical- or phonological-level difficulty (Fraundorf & Watson, 2014), prior to new information (Barr & Seyfeddinipur, 2010) and prior to low frequency words and words with lower cloze probability (Beattie & Butterworth, 1979; Dammalapati, Rajkumar, Ranjan, & Agarwal, 2021; Goldman-Eisler, 1958a, 1958b). In addition, information spoken after a disfluency is better recalled and is responded to faster (Collard, Corley, MacGregor, & Donaldson, 2008; Corley, MacGregor, & Donaldson, 2007; Corley & Hartsuiker, 2011; Diachek & Brown-Schmidt, 2023; Fraundorf & Watson, 2011; MacGregor, Corley, & Donaldson, 2010). Some accounts explain these benefits with the longer processing time associated with the temporal delay for word processing given by disfluencies (Bailey & Ferreira, 2003; Corley et al., 2007; Corley & Hartsuiker, 2011; MacGregor et al., 2010). Other accounts propose that hesitations cue the listener's attention to upcoming speech and facilitate processing (Collard et al., 2008; Fraundorf & Watson, 2014). Some accounts also propose that disfluencies may inform lexical processing, by updating listeners' predictions about upcoming speech. Eye-tracking studies show that listeners' fixations are directed to less accessible (e.g. lower frequency) items than more accessible ones following disfluencies (Arnold, Tanenhaus, Altmann, & Fagnano, 2004,

2007; Bosker, Quené, Sanders, & de Jong, 2014). An eye-tracking study in virtual reality found that repairs reduced anticipatory fixations toward a predicted referent and that more fixations were directed towards the speaker upon hearing hesitations (Huizeling, Peeters, & Hagoort, 2022). EEG evidence also suggests that the N400 response to unexpected words is reduced following both filled and silent disfluencies (Corley et al., 2007; MacGregor et al., 2010), but another EEG study found no evidence for a reduction in the N400 or prediction updating but found results consistent with the allocation of more processing resources to the incoming input (Huizeling, Peeters, & Hagoort, 2026).

Many studies that investigated the effects of disfluencies on speech processing have been run as constrained experiments with eye-tracking and EEG. The effect of disfluencies in naturalistic speech from a listener's perspective has instead received less attention. The advantage of using naturalistic stimuli is that the probability distributions of the input in spontaneous and narrative speech are more likely to match participants' linguistic experiences than isolated experimental sentences (Jaeger, 2010). It is therefore important to assess whether disfluencies have an influence on lexical processing also in naturalistic speech. An EEG study on the comprehension of spontaneous speech found that disfluencies themselves elicited a weaker speech-tracking response than both content and function words, suggesting that they are not interpreted linguistically and may be processed more like pauses than words (Agmon, Jaeger, Tsarfaty, Bleichner, & Zion Columbic, 2023). In this study, therefore, we asked whether and how disfluencies affected the neural response to words following them. In particular, we focused on the effect of disfluencies on variables indexing word processing effort (i.e. word frequency and surprisal).

The neural response to naturalistic language processing has been investigated in many neuroimaging studies (e.g. Bhattasali et al., 2019; Brennan, Stabler, Van Wagenen, Luh, & Hale, 2016; Coopmans et al., 2025; Giglio, Ostarek, et al., 2024; Hausfeld, Hamers, & Formisano, 2024; Wehbe et al., 2014). In fMRI, word processing effort metrics for each word (e.g. word frequency, word surprisal, or syntactic processing load) are convolved with the hemodynamic response function (HRF) to derive expected time courses of brain activity for the incoming speech which are then compared to the measured neural time courses (Brennan, 2016). Even if fMRI does not allow for modelling responses to each word, in contrast to EEG or MEG, the convolution of word-by-word metrics with the HRF generates slow varying time courses that predict differences in processing effort over time relative to different metrics. Left fronto-temporal areas of the brain respond to the incoming linguistic input as a function of these word-by-word processing metrics. In particular, previous studies found that lower predictability of a word based on preceding context (word surprisal) is associated with increases in brain activity in the left and right temporal lobe and in the left inferior frontal gyrus (Bhattasali & Resnik, 2021; Lopopolo, Frank, Bosch, & Willems, 2017; Russo et al., 2020; Willems, Frank, Nijhof, Hagoort, & van den Bosch, 2016). When the analysis is run with a functional localization approach the effects of surprisal are visible throughout the language network in temporal and frontal regions (Shain, Blank, van Schijndel,

Schuler, & Fedorenko, 2020). Lower word frequency is associated with activity increases in the posterior temporal lobe and pars orbitalis of the left inferior frontal gyrus (LIFG) (Shain et al., 2020). Studies with higher temporal resolution using MEG find similar effects for the response to word frequency and surprisal (Armeni et al., 2019; Heilbron et al., 2022; Huizeling, Arana, Hagoort, & Schoffelen, 2022; Slaats, Weissbart, Schoffelen, Meyer, & Martin, 2023).

Here, we applied this word-by-word incremental approach to the study of naturalistic language processing to understand whether the presence of disfluencies affected the word-by-word neural responses to indices of word processing load. We used linguistic stimuli that were recorded from a speaker spontaneously recalling the events of two television series. The speech included silent pauses as well as filled hesitations. We modelled the incremental neural response to words based on word length, frequency, entropy and surprisal, and position of the word in the sentence and in text. We then asked whether words preceded by a disfluency were processed differently from words spoken fluently (i.e. not preceded by disfluencies). In addition, we investigated whether the response to word frequency and surprisal varied as a function of the presence or absence of disfluencies. We therefore asked whether disfluencies affect listeners' lexical processing in an fMRI study of naturalistic speech. Although fMRI is less sensitive to fast temporal dynamics than methods with higher temporal resolution, it has been used to successfully investigate incremental processing in previous studies. The use of fMRI to study naturalistic and disfluent speech can additionally uncover which brain areas show sensitivity to the presence of disfluencies. Previous work found that listeners anticipate hearing a lower frequency or less predictable word after hearing a disfluency (Arnold et al., 2004; Corley et al., 2007). These findings suggest that the presence of a hesitation disfluency is informative for participants, potentially preparing them to hear a lower frequency, less predictable or more surprising word following a hesitation disfluency, either by leading to an update in their predictions or to an increase in attention (e.g. Collard et al., 2008). Here we tested the extent that the effects of frequency and surprisal on word processing are influenced by the presence of a preceding hesitation. First, we looked at silent pauses and filled hesitations together. In a follow-up analysis, we asked whether the effects were stable focusing on silent pauses and filled disfluencies separately, since different types of disfluencies can differ in their discourse functions (e.g., Clark & Fox Tree, 2002; Diskin, 2017; Fox Tree, 2001; Fraundorf & Watson, 2014).

## 2. Methods

### 2.1. Participants

This study was based on a dataset collected by Zadbood, Chen, Leong, Norman, and Hasson (2017, 2018) and shared on OpenNeuro (<https://openneuro.org/datasets/ds001110/versions/>

00003), after participants provided informed written consent before the start of the study in accordance with experimental procedures approved by the Princeton University Institutional Review Board. The original collected dataset included 52 right-handed native English speakers (age 18–45). Fifteen participants were excluded because of head motion ( $n = 4$ ), for falling asleep ( $n = 4$ ), due to poor memory in a post-scan free recall test ( $n = 5$ ), or for having seen the movie before ( $n = 2$ ). As a result, the data for 36 participants were shared and analyzed here.

### 2.2. Stimuli

The audio recordings for the listening task were produced by a participant of a separate experiment (Chen et al., 2017, 2018). This participant was asked to verbally describe the events of the first episode of BBC television series *Merlin* and *Sherlock* while undergoing fMRI scanning. After watching each episode, they were immediately instructed “to describe what they recalled of the movie in as much detail as they could, to try to recount events in the original order they were viewed in, and to speak for at least 10 min if possible but that longer was better. They were told that completeness and detail were more important than temporal order, and that if at any point they realized they had missed something, to return to it. Participants were then allowed to speak for as long as they wished, and verbally indicated when they were finished (for example, “I’m done”). During this session they were presented with a static black screen with a central white dot (but were not asked to, and did not, fixate).” (Chen et al., 2017). Their speech was recorded in the scanner with an MR-compatible microphone. This participant produced both a recall of *Merlin* and of *Sherlock*. The audio recall for *Merlin* was 14.7 min long and included 2141 words. The audio recall for *Sherlock* was 17.5 min long and included 2468 words.

### 2.3. Linguistic annotations

The *Sherlock* and *Merlin* recalls were transcribed and time-stamped using gentle ([github.com/lowerquality/gentle](https://github.com/lowerquality/gentle), see also Born et al., 2023). Sentence boundaries were manually annotated based on the syntactic structure and the recordings, which were informative thanks to the presence of prosody and pauses. A new sentence would start whenever the syntactic structure was interrupted in some way (excluding fillers), for example by a restart, a parenthetical insertion, a repetition, etc. Coordinated clauses were also divided into separate sentences. For additional discussion on the sentence boundaries annotated here and in general, see (Agmon, Cho, et al., 2025; Agmon et al., 2024; Agmon, Jaeger, et al., 2025; Giglio, Ostarek, et al., 2024). The text was manually annotated for the presence of disfluencies. Disfluencies were defined as silent pauses longer than 250 msec (Goldman-Eisler, 1968), as filled pauses such as “uh” and “uhm”, and “like” only when used as a filler, since it has been found to signal vagueness and uncertainty in the upcoming speech (Andersen, 1998; Jucker, Smith, & Lüdge, 2003). We excluded discourse markers like “you know”, “I mean”, etc., because these are longer and have a more specific function in discourse (e.g. Fox Tree, 2007; 2010;

Holmes, 1990). We initially analyzed silent and filled disfluencies together, to optimize our power to estimate the response to disfluencies with the low temporal resolution of fMRI (which does not allow for focusing on single-word responses when presented within discourse). For the purposes of the analysis, we marked each word as either preceded or not preceded by a disfluency (1 vs –1). In a follow-up analysis, we looked at the effects of silent pauses and filled disfluencies separately. We additionally modelled word processing effort with a few baseline variables: number of syllables, log frequency from SUBTLEX-US (Brysbaert, New, & Keuleers, 2012), word position in the sentence and in text, and entropy. The baseline variables were included to model effects of word processing effort and context that were not specific to word predictability measured with surprisal. In the fMRI analysis, the negative log of word frequency was used, so that higher values corresponded to lower frequency words, making the units of frequency comparable to those of surprisal and simplifying the reporting of results.

We quantified both word entropy and surprisal from transformer model GPT-2 (Radford et al., 2019). Filled disfluencies were removed from the transcripts used for computing GPT-2 surprisal and entropy. We used GPT-2 XL via the TensorFlow implementation provided by HuggingFace's Transformers package (Wolf et al., 2020). GPT-2 allows for a context of up to 1024 tokens (each token corresponds to a word or a part of a word), which is shorter than the transcripts used here. For the first 1024 tokens in each transcript, the context consisted of all tokens available until then. For tokens following the 1024th token, we used a window approach, with a context of at least 700 tokens for each window (increasing for the later tokens in the window up to the 1024th token). For example, the surprisal for the fifth token in the recall was calculated on a context of four tokens, while the surprisal for the 1100th token was based on a context of approximately 800 tokens. A window approach was less computationally expensive than a maximal context of 1024 tokens for each word while still providing a large context on which to estimate surprisal. Surprisal was calculated as the negative logarithm of the conditional probability of the word based on context. Entropy was calculated as the sum of the conditional probability of each next possible word, weighted by the negative logarithm of this probability.

The distributions of word length, word frequency, word surprisal, entropy and word position in the sentence and in text for words preceded versus not preceded by a disfluency are presented in Figs. 1 and 2. We compared word characteristics for words preceded versus not preceded by a disfluency using linear mixed-effects models with *lme4* in R (Bates, Mächler, Bolker, & Walker, 2015). Each model included disfluency as a fixed effect, coded with a Helmert contrast comparing disfluency versus no disfluency and silent versus filled disfluencies. We looked at pairwise comparisons using *emmeans* (Lenth et al., 2022). We additionally included random intercepts for movie (Sherlock versus Merlin). Significance for each contrast was determined using *lmerTest* (Kuznetsova, Brockhoff, Christensen, & Jensen, 2020). The distribution of disfluencies relative to the grammatical class of words is also presented in Supplementary Figures 1 and 2.

## 2.4. Procedure

In this experiment, participants listened to the audio recording of a person recalling the events of an episode of the BBC television series *Merlin* or *Sherlock*, while undergoing fMRI scanning. In the current study, we focused on the listening part of the experiment. Among analyzed participants, 18 listened to the recall of *Merlin*, and 18 listened to the recall of *Sherlock*. Participants were instructed to listen to the stimuli carefully and were told that there would be a memory test after the experiment. The memory test consisted of a free recall test in which participants were asked to write down the events they remembered in as much detail as possible. Each participant was given a memory score by three independent raters, and participants with a score 1.5 SD below the mean were excluded ( $n = 5$ , see Participants).

## 2.5. fMRI data acquisition and analysis

### 2.5.1. Data acquisition

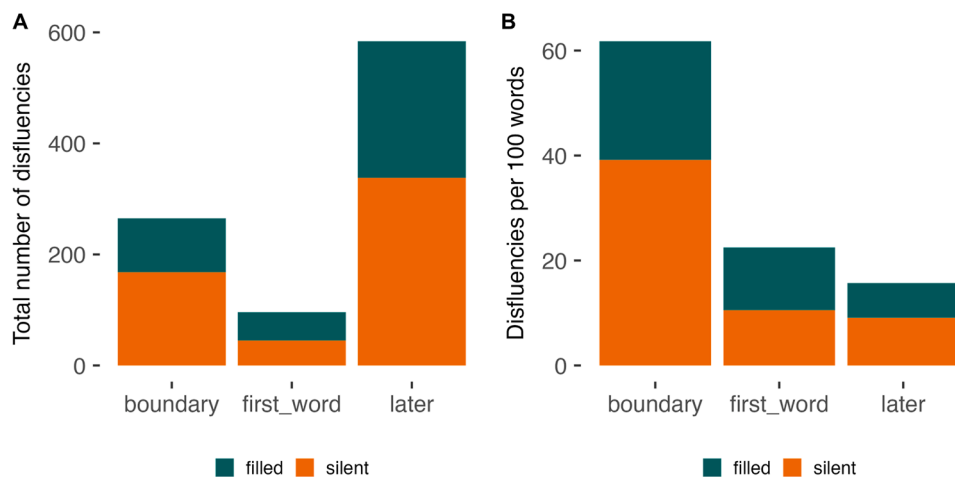
The data were collected on a 3T full-body scanner (Siemens Skyra) with a 20-channel head coil. Functional images were acquired using a T2\*-weighted echo planar imaging pulse sequence (TR 1500 msec, TE 28 msec, flip angle 64, whole-brain coverage 27 slices of 4 mm thickness, in-plane resolution  $3 \times 3 \text{ mm}^2$ , FOV  $192 \times 192 \text{ mm}^2$ ). Anatomical images were acquired using a T1-weighted MPRAGE pulse sequence (.89 mm<sup>3</sup> resolution).

### 2.5.2. fMRI preprocessing

Preprocessing was performed on the raw data shared on OpenNeuro (Zadbood, Chen, Leong, Norman, & Hasson, 2018) using *fMRIPrep* 20.2.6 (Esteban et al., 2018), which is based on *Nipype* 1.7.0 (Gorgolewski et al., 2011, 2018). The following text in this section refers to standard *fMRIPrep* preprocessing steps. The T1-weighted (T1w) image was corrected for intensity non-uniformity (INU) with *N4BiasFieldCorrection* (Tustison et al., 2010), distributed with ANTs 2.3.3 (Avants, Epstein, Grossman, & Gee, 2008), and used as T1w-reference throughout the workflow. The T1w-reference was then skull-stripped with a *Nipype* implementation of the *ants-BrainExtraction.sh* workflow (from ANTs), using *OASIS30ANTs* as target template. Brain tissue segmentation of cerebrospinal fluid (CSF), white-matter (WM) and gray-matter (GM) was performed on the brain-extracted T1w using *fast* (FSL 5.0.9 (Zhang, Brady, & Smith, 2001)). Brain surfaces were reconstructed using *recon-all* (FreeSurfer 6.0.1 (Dale, Fischl, & Sereno, 1999)), and the brain mask estimated previously was refined with a custom variation of the method to reconcile ANTs-derived and FreeSurfer-derived segmentations of the cortical gray-matter of *Mindboggle* (Klein et al., 2017). For each BOLD run, the following preprocessing was performed. First, a reference volume and its skull-stripped version were generated using a custom methodology of *fMRIPrep*. Susceptibility distortion correction (SDC) was omitted, because no fieldmap was acquired. The BOLD reference was then co-registered to the T1w reference using *bbregister* (FreeSurfer) which implements boundary-based registration (Greve & Fischl, 2009). Co-registration was configured with six degrees of freedom. Head-motion parameters with respect to the BOLD reference



**Fig. 1 – A:** Density plots showing the density of each set of words (preceded by silent pauses, filled disfluencies and no disfluencies) relative to all word characteristics modelled here. The x-axis refers to the title of each figure, while the y-axis shows the density. **B:** Bar plots show mean  $\pm$  standard error of the mean for each word characteristic. The y-axis refers to the title of the plot. \* indicates  $p < .05$ .



**Fig. 2 – A:** Total absolute number of disfluencies produced before the first word in a sentence (i.e. at a sentence boundary, out of 429 total words produced in that position), before the second word in a sentence (out of 427 words in that position), or later in the sentence (out of 3719 words in that position). **B:** Percentage of first, second and later words in a sentence that were preceded by silent and filled disfluencies.

(transformation matrices, and six corresponding rotation and translation parameters) are estimated before any spatiotemporal filtering using mcflirt (FSL 5.0.9 (Jenkinson, Bannister, Brady, & Smith, 2002)). The BOLD time-series (including slice-timing correction when applied) were resampled onto their original, native space by applying the transforms to

correct for head-motion. The BOLD time-series were resampled into standard space, generating a *preprocessed BOLD run* in MNI152Nlin2009cAsym space. First, a reference volume and its skull-stripped version were generated using a custom methodology of fMRIprep. Automatic removal of motion artifacts using independent component analysis (ICA-AROMA (Pruim

et al., 2015)) was performed on the *preprocessed BOLD* on MNI space time-series after removal of non-steady state volumes and spatial smoothing with an isotropic, Gaussian kernel of 6 mm FWHM (full-width half-maximum). The “aggressive” noise-regressors were collected and placed in the corresponding confounds file. Confounding time-series were calculated based on the *preprocessed BOLD* for framewise displacement (FD), and the derivative of the relative (frame-to-frame) bulk head motion variance (DVARs). FD was computed using two formulations following Power (absolute sum of relative motions (Power et al., 2014) and Jenkinson (relative root mean square displacement between affines (Jenkinson et al., 2002). FD and DVARs are calculated for each functional run, both using their implementations in Nipype (following the definitions by (Power et al., 2014). Additionally, a set of physiological regressors were extracted to allow for component-based noise correction (CompCor (Behzadi, Restom, Liau, & Liu, 2007)). Principal components are estimated after high-pass filtering the *preprocessed BOLD* time-series (using a discrete cosine filter with 128s cut-off) for anatomical CompCor (aCompCor). For aCompCor, three probabilistic masks (cerebrospinal fluid (CSF), white matter (WM) and combined CSF + WM) are generated in anatomical space.

### 2.5.3. fMRI data analysis

The *preprocessed BOLD* images in MNI standard space were smoothed with 5 mm FWHM Gaussian kernel in SPM12 (version 7771, Penny, Friston, Ashburner, Kiebel, & Nichols, 2011) in Matlab 2023a. The first-level design matrix included as regressors of interest the onset of each word, as well as parametric modulations detailed below to investigate effects of disfluency and surprisal on word processing. Temporal derivatives were also estimated but not further inspected. As nuisance regressors we included DVARs, framewise displacement, 5 aCompCor parameters, 6 motion parameters and the AROMA noise components to optimally account for motion-related and physiological noise. Following parameter estimation, we extracted contrast images for each participant for the word condition and its parametric modulations. The contrast images were then entered into the group level design matrix with one-sample T-tests. Results were extracted with a voxel-level threshold at  $p < .001$  uncorrected, and a  $p < .05$  family-wise error correction as a cluster threshold.

Parametric modulations allow for modelling continuous variations to the amplitude of the response of a condition of interest, in this case the word regressor. In SPM, the order of the parametric modulations can affect the results, since each additional modulation is orthogonalized with respect to the previous ones. We thus used different models to investigate the effects of different variables. All models included baseline parametric modulations for number of syllables as an index for length, for log frequency, for word position in the sentence, for word position in text and for entropy in this order, with increasing importance of context for each regressor. To look at the effect of disfluencies, we additionally added parametric modulations for surprisal and disfluency in this order. To look at the effect of surprisal we included disfluency first and then surprisal. To focus on the effect of disfluencies on surprisal and frequency, we additionally added a parametric

modulation for the interaction between disfluency and surprisal/frequency as the multiplication between these two variables, in two separate models for the interaction with surprisal and frequency. In a follow-up analysis, we also looked for the effects of silent and filled disfluencies separately, using the same approach just mentioned, but marking disfluencies as either silent or filled. All modulations were mean-centered. The analysis was run on the whole-brain for all modulations, including interactions. We preferred not to constrain our search to regions that were found to be involved in the surprisal and frequency effects, because any clusters involved in an interaction may not have appeared as main effects.

## 3. Results

### 3.1. Disfluency distribution in speech

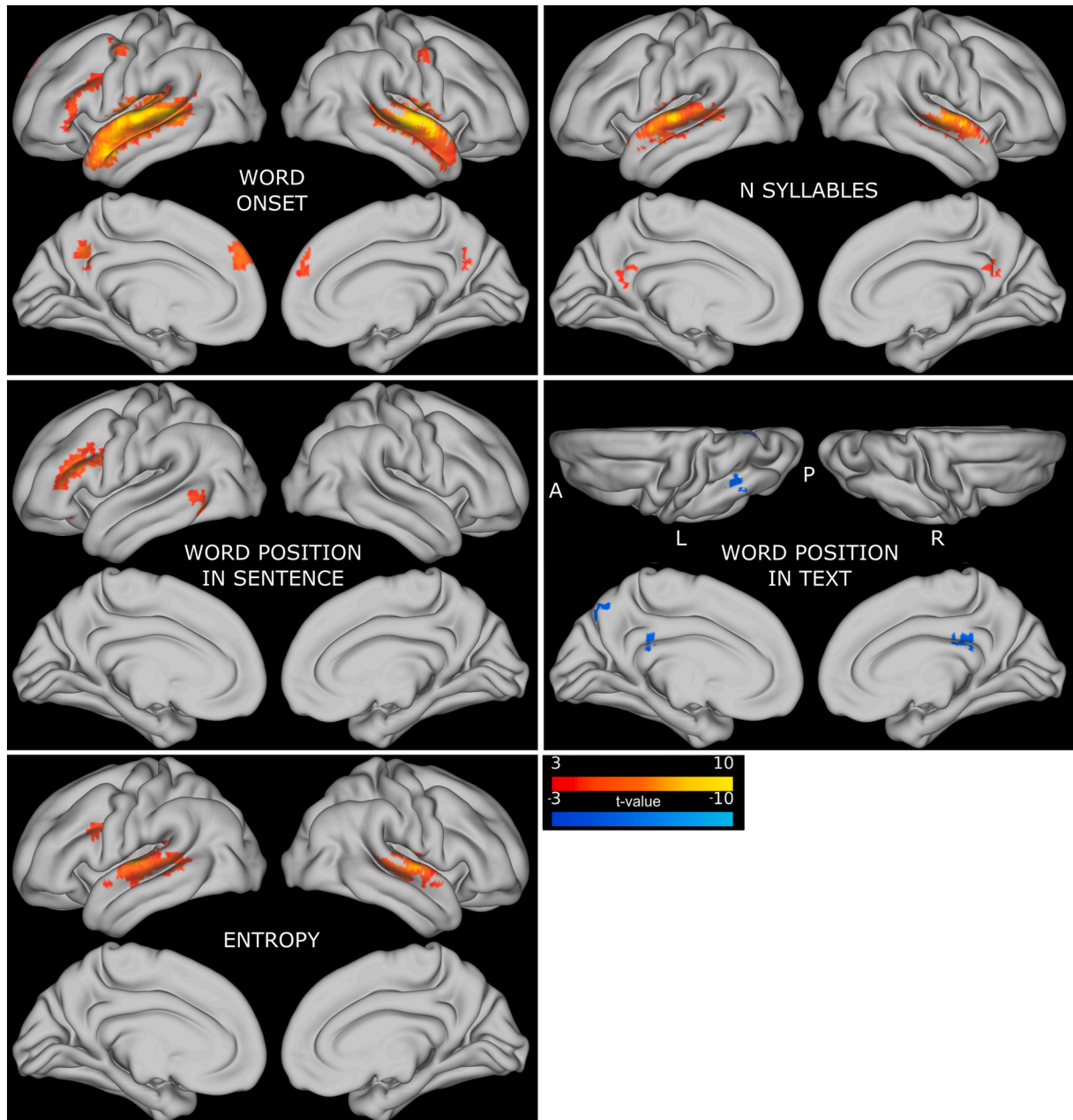
Overall, 15% of words were preceded by a silent pause and 10.8% of words were preceded by a filled disfluency. When including filled disfluencies among the count of words, filled disfluencies were 9.8% of all words. Disfluencies often occurred before coordinating conjunctions, determiners, existential *there*, preposition or subordinating conjunctions, noun phrases, pronouns, adverbs, interjections, and verbs (Supplementary Figs. 1–2). Disfluencies occurred more often at sentence boundaries and earlier in a sentence (Figs. 1–2). Fig. 2 plots both the total number of disfluencies throughout the sentence in the whole speech (Fig. 2A), and the percentage of first, second and later words that were preceded by either type of disfluency (Fig. 2B). Words preceded by silent disfluencies were shorter than words preceded by filled disfluencies as well as words not preceded by disfluencies (silent versus filled: estimate = .12, SE = .04,  $t = 2.86$ ,  $p = .012$ ; silent versus no disfluency: estimate = .084, SE = .03,  $t = 2.83$ ,  $p = .013$ ). Word frequency was also higher for words preceded by silent disfluencies relative to filled ones and no disfluencies (silent versus filled: estimate = .37, SE = .097,  $t = 3.83$ ,  $p < .001$ ; silent versus no disfluency: estimate = .25, SE = .07,  $t = 3.68$ ,  $p < .001$ ). Both filled and silent disfluencies also occurred earlier in a sentence (filled versus no disfluency: estimate = 2.57, SE = .33,  $t = 7.73$ ,  $p < .001$ ; silent versus no disfluency: estimate = 2.02, SE = .3,  $t = 7.5$ ,  $p < .001$ ), while there was no difference in the distribution of disfluencies throughout the whole speech. Entropy was higher for words preceded versus not preceded by a disfluency, and higher for filled than silent disfluencies (silent versus no disfluency: estimate = .29, SE = .08,  $t = 3.43$ ,  $p = .002$ ; silent versus filled: estimate = .58, SE = .12,  $t = 4.78$ ,  $p < .0001$ ; filled vs no disfluency: estimate = .87, SE = .098,  $t = 8.8$ ,  $p < .001$ ). Surprisal was also higher for words preceded by a disfluency, and for filled versus silent disfluencies (silent vs. no disfluency: estimate = .91, SE = .17,  $t = 5.38$ ,  $p < .001$ ; silent versus filled: estimate = 1.33, SE = .25,  $t = 5.4$ ,  $p < .001$ ; filled versus no disfluency: estimate = 2.24, SE = .2,  $t = 11.3$ ,  $p < .001$ ). Therefore, disfluencies occurred in conditions of higher planning requirements, as observed by previous aforementioned studies (Beattie & Butterworth, 1979; Clark & Fox Tree, 2002; Dammalapati et al., 2021; Fraundorf &

Watson, 2014): earlier in a sentence, before words with higher entropy and higher surprisal. The only exception were silent disfluencies that occurred more often before shorter and more frequent words.

### 3.2. fMRI baseline results

First, we inspected the baseline word predictors (word onset, length, word position in the sentence and in text, and entropy) (Fig. 3, Table 1). The BOLD response to words was most strongly localized to the left and right superior temporal

cortex. Additional clusters were found in the left IFG pars triangularis, left and right precentral gyrus, left medial superior frontal cortex, right superior temporal sulcus (STS), right cerebellum (lobule VI). Longer words additionally elicited stronger responses in the left and right superior temporal gyrus (STG), and in the right posterior cingulate cortex. Later word position in the sentence was associated with stronger responses in the left IFG, pars triangularis and orbitalis, and in the left inferior temporal gyrus. Later word position across all text was associated with a decrease in the left angular gyrus (AG), left precuneus and right middle cingulate cortex. Words



**Fig. 3 – Whole-brain results for the baseline word predictors. Results are significant with  $p < .05$  FWE cluster correction. Warm colors indicate positive activity (where the response to word processing increases), cold colors indicate a decrease in activity (relative to word processing). The color bar refers to voxel t values.**

**Table 1 – fMRI whole-brain cluster and peak statistics for baseline processing variables.**

| Contrast             | Cluster           |      | Peak voxel (MNI coordinates) |     |     |     | Anatomical location            |
|----------------------|-------------------|------|------------------------------|-----|-----|-----|--------------------------------|
|                      | p (FWE corrected) | size | t value                      | x   | y   | z   |                                |
| Word onset           | 0                 | 719  | 13.68                        | 60  | –10 | –2  | RSTG                           |
|                      | 0                 | 1039 | 13.43                        | –52 | –16 | 10  | LSTG                           |
|                      | .001              | 43   | 6.05                         | –54 | –4  | 54  | Lprecentral gyrus              |
|                      | .045              | 20   | 5.89                         | 18  | –72 | –26 | R Cerebellum (VI lobule)       |
|                      | 0                 | 96   | 5.74                         | 0   | 54  | 38  | L Medial SFG                   |
|                      | 0                 | 46   | 5.57                         | –4  | –54 | 38  | L Precuneus                    |
|                      | 0                 | 159  | 4.74                         | –54 | 26  | 6   | LIFGtri                        |
|                      | .045              | 20   | 4.31                         | 56  | –4  | 46  | Rprecentral gyrus              |
| N syllables          | 0                 | 429  | 9.22                         | –64 | –12 | 6   | LSTG                           |
|                      | 0                 | 274  | 8.49                         | 62  | –4  | –2  | RSTG                           |
| Position in sentence | 0                 | 366  | 7.83                         | –48 | 30  | 18  | LIFGtri                        |
|                      | 0                 | 86   | 5.6                          | –48 | –58 | –10 | LInferior Ttemporal Gyrus      |
|                      | .013              | 26   | 5.05                         | –36 | 36  | –14 | LIFGorb                        |
| Position in text     | .001              | 38   | –4.61                        | –36 | –60 | 38  | LAG                            |
|                      | .003              | 33   | –4.43                        | 2   | –34 | 26  | Posterior/mid cingulate cortex |
|                      | 0                 | 44   | –4.37                        | –6  | –76 | 46  | L Precuneus                    |
|                      | .035              | 20   | –4.16                        | –4  | –28 | –2  | L Thalamus                     |
| Entropy              | 0                 | 377  | 7.99                         | –54 | –22 | 6   | LSTG, LMTG                     |
|                      | 0                 | 223  | 7.57                         | 66  | –6  | 2   | RSTG                           |
|                      | 0                 | 52   | 5.14                         | –40 | 12  | 26  | LIFG                           |

with higher entropy also had a stronger response in the left and right STG, as well as in the left IFG pars opercularis. The BOLD response to word processing effort, therefore, was localized to fronto-temporal regions most strongly in the left hemisphere.

### 3.3. Disfluencies and the interaction with surprisal and frequency

Disfluencies, word frequency and word surprisal also affected the BOLD response to words (Fig. 4, Table 2). Words preceded by a disfluency were associated with an activity increase in the left and right STG and STS. There was also a smaller cluster in the right supramarginal gyrus (SMG). There was a weaker BOLD response in the right posterior temporal cortex to words that were preceded by a disfluency relative to those that were spoken fluently. Increased word surprisal elicited a greater BOLD increase in the left STG and right STS, and a greater decrease in BOLD in the left and right AG, and in the left middle occipital cortex. There was also an interaction between disfluencies and surprisal (Fig. 4). The effect of surprisal decreased for words preceded by a disfluency in a cluster in the left posterior middle temporal gyrus (MTG)/AG, not overlapping with the disfluency and surprisal clusters.

The effect of word frequency was associated with stronger responses in the left STS, left middle frontal gyrus (MFG) and left inferior temporal gyrus (lower frequency words elicited higher activity). Disfluencies also interacted with word frequency. The effect of frequency increased for words preceded by a disfluency in the left STG, left IFG (pars triangularis and opercularis), left posterior MTG and left SMG. Therefore, disfluencies enhanced the brain response to word frequency in fronto-temporal areas.

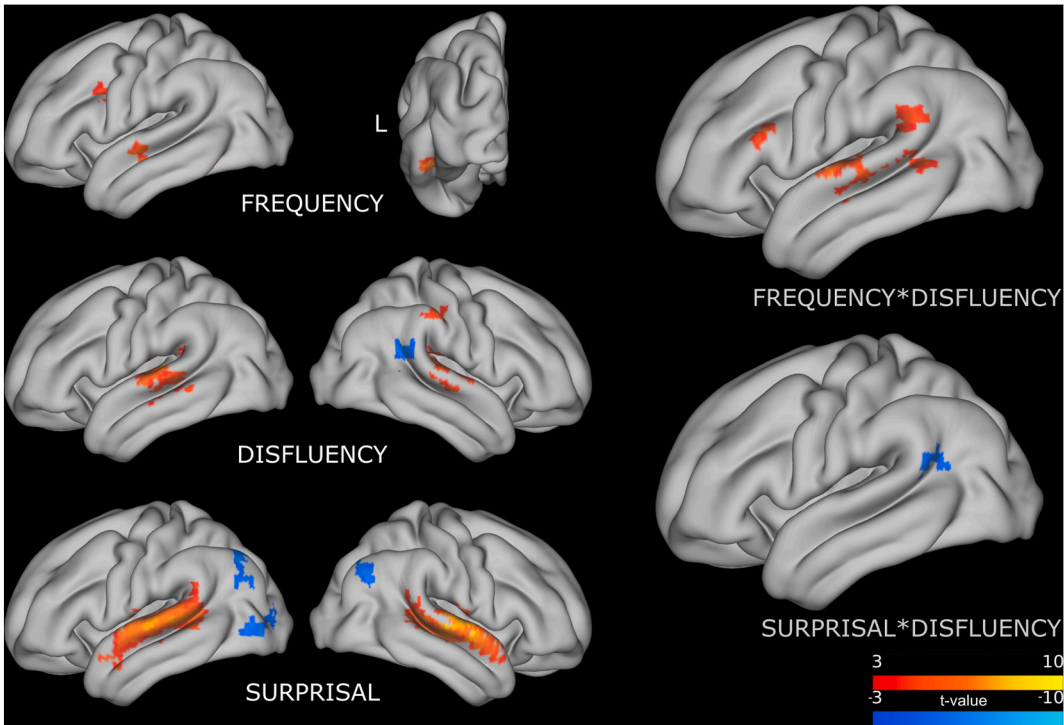
### 3.4. Silent and filled disfluencies

Our analysis so far focused on disfluencies in general, including both silent and filled disfluencies. However, previous studies have shown that their function in discourse can differ. We therefore also ran the same analysis focusing on each disfluency type separately, to understand whether the interactions we found were driven by either type of disfluency or if they had similar impacts on brain activity, with the caveat that both analyses would have less power than the grouped one (Fig. 5, Table 3).

Words preceded by silent disfluencies were associated with an activity increase in a cluster in the LSTG and RSTG that closely mirrored the cluster found for all types of disfluencies. There were increases in activity also in the LIFG pars opercularis, RSMG, another cluster in the left inferior parietal lobule, and the left superior medial frontal gyrus (SFG). There was an activity decrease in the right AG. Words preceded by filled disfluencies showed a similar activity increase in a cluster in the left and right STG, which however did not extend as far along the gyrus. Activity decreased in the left SMG and right AG.

The interaction between silent disfluencies and surprisal elicited an activity increase in a cluster in the left STG, which was also found to be modulated by both disfluencies and surprisal. The interaction between silent disfluencies and frequency was associated with activity increases in the LSTG and LMTG, LIFG pars triangularis and opercularis, and LSMG. This pattern closely resembled the interaction found with both types of disfluencies.

The interaction between filled disfluencies and surprisal did not lead to any significant activation with FWE cluster correction, but it was associated with activity decreases in the LSTG, left AG, LMTG and right thalamus with  $p < .05$ .



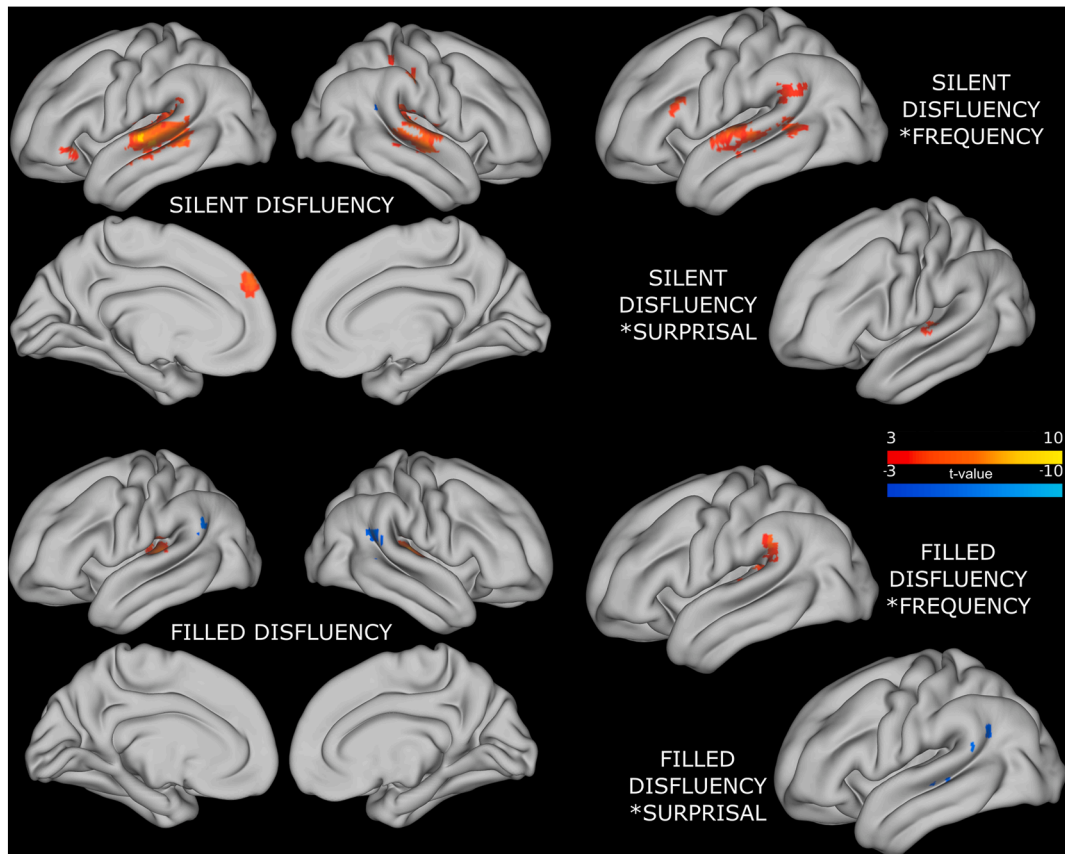
**Fig. 4 – Whole-brain results for frequency, disfluency, surprisal and their interactions. Results are significant with  $p < .05$  FWE cluster correction. Warm colors indicate positive activity (where the response to word processing increases), cold colors indicate a decrease in activity (relative to word processing). The color bar refers to voxel t values.**

**Table 2 – fMRI whole-brain cluster and peak statistics for variables for interest.**

| Contrast                   | Cluster             |      | Peak voxel (MNI coordinates) |     |     |     | Anatomical location  |
|----------------------------|---------------------|------|------------------------------|-----|-----|-----|----------------------|
|                            | $p$ (FWE corrected) | size | t value                      | x   | y   | z   |                      |
| FREQUENCY                  | .004                | 31   | 7.05                         | −46 | −52 | −14 | LITG                 |
|                            | .019                | 23   | 5.46                         | −60 | −12 | −6  | LMTG                 |
|                            | 0                   | 48   | 5.04                         | −46 | 6   | 34  | LMFG                 |
| Surprisal                  | 0                   | 455  | 9.16                         | 62  | −10 | 6   | RSTG                 |
|                            | 0                   | 471  | 8.58                         | −60 | −16 | 2   | LSTG                 |
|                            | 0                   | 98   | −6.11                        | −34 | −70 | 42  | LAG                  |
|                            | 0                   | 47   | −5.15                        | −48 | −76 | −2  | L Mid occipital lobe |
|                            | 0                   | 61   | −4.64                        | 44  | −64 | 38  | RAG                  |
| Disfluency                 | 0                   | 201  | 6.25                         | −64 | −24 | 10  | LSTG, LMTG           |
|                            | 0                   | 156  | 5.89                         | 56  | −16 | 6   | RSTG                 |
|                            | .001                | 41   | 5.55                         | 60  | −28 | 42  | RSMG                 |
|                            | 0                   | 81   | −6.07                        | 48  | −42 | 22  | RSTG                 |
| Interaction with surprisal | 0                   | 45   | −4.26                        | −58 | −52 | 18  | LMTG                 |
| Interaction with frequency | 0                   | 132  | 6.48                         | −58 | −16 | 2   | LSTG                 |
|                            | 0                   | 57   | 5.65                         | −54 | 18  | 26  | LIFGtri, LIFGoper    |
|                            | .003                | 33   | 4.61                         | −54 | −48 | 2   | LMTG                 |
|                            | .018                | 23   | 4.4                          | −58 | −42 | 26  | LSMG                 |

uncorrected. The left AG cluster in particular mirrors the interaction found with both types of disfluencies. The interaction between filled disfluencies and frequency was significantly associated with an activity increase in the LSMG and in the LSTG ( $p < .05$  FWE corrected), which partly overlapped with the interaction with both types of disfluencies.

Overall, silent and filled disfluencies modulated the activation of words following them in a similar way, by leading to stronger activation in the superior temporal cortex. In addition, both silent and filled disfluencies enhanced the word frequency effect in the LSTG, as also found when considering both types of disfluencies together. Instead, silent and filled



**Fig. 5 – Whole-brain results for silent and filled disfluencies separately, as well as their interactions with word frequency and surprisal. Results are significant with  $p < .05$  FWE cluster correction in all cases except for the interaction between filled disfluency and surprisal ( $p < .001$  at the voxel level,  $p < .05$  uncorrected at the cluster level). Warm colors indicate positive activity (where the response to word processing increases), cold colors indicate a decrease in activity (relative to word processing). The color bar refers to voxel t values.**

disfluencies dissociated in how they affected the effect of surprisal. Silent disfluencies led to a stronger response to surprisal in the LSTG, while filled disfluencies appeared to lead to an activity decrease from the left AG along the LSTS.

#### 4. Discussion

In an fMRI study of narrative comprehension, we investigated whether disfluencies affect the online processing of words in discourse. Participants listened to the free recall of a story that was unrehearsed and thus spontaneously spoken. We modelled the brain response to discourse by quantifying word processing effort using baseline word characteristics like word length, word position in a sentence and entropy. Critically, we asked how the brain response to words changed following disfluencies, and whether the presence of disfluencies affected the response to word predictability in context (measured with surprisal) and word frequency. We found that word frequency was associated with increased brain activity in left temporal and frontal clusters, where lower frequency words elicited higher activation. Increased word surprisal elicited larger responses in the superior

temporal lobe. Words following a disfluency were also associated with higher brain activity in the superior temporal gyrus. Clusters in the left superior temporal gyrus and inferior frontal gyrus showed an enhanced response to word frequency following disfluencies, while a more posterior temporal cluster showed a reduction in the response to surprisal following disfluencies. A follow-up analysis on silent and filled disfluencies separately found many similarities in how silent and filled disfluencies affected word processing, but they dissociated in how they affected the response to word surprisal.

The presence of interactions between disfluencies and word frequency and surprisal in BOLD activity, therefore, suggests that disfluencies affect incremental word comprehension. Disfluencies themselves elicited stronger responses to the following word in an area of the superior temporal gyrus that was also affected by word surprisal. Similar clusters were found in previous fMRI studies that focused on the response to disfluencies versus fluent speech (Eklund & Ingvar, 2016; Smirnova et al., 2020). The presence of disfluencies in speech is thought to drive an increase in attention and processing resources (Collard et al., 2008). This increase in attention could be driven by top-down processes preparing

**Table 3 – fMRI whole-brain cluster and peak statistics for the effects of silent and filled disfluencies analyzed separately.**

| Contrast                          | Cluster            |      | Peak voxel (MNI coordinates) |     |     |    | Anatomical location       |
|-----------------------------------|--------------------|------|------------------------------|-----|-----|----|---------------------------|
|                                   | p (FWE corrected)  | size | t value                      | x   | y   | z  |                           |
| Silent disfluencies               | 0                  | 443  | 8.3                          | −64 | 16  | 2  | LSTG                      |
|                                   | 0                  | 326  | 6.98                         | 66  | −22 | 6  | RSTG                      |
|                                   | .007               | 28   | 5.46                         | 56  | −24 | 38 | RSMG                      |
|                                   | 0                  | 46   | 5.32                         | −4  | 54  | 38 | Left SFG medial           |
|                                   | .016               | 24   | 4.32                         | 30  | −40 | 50 | RInferior Parietal Lobule |
|                                   | .037               | 20   | 4.26                         | −46 | 26  | −6 | LIFG pars orbitalis       |
|                                   | .016               | 24   | −4.67                        | 48  | −46 | 22 | RAG                       |
| Silent disfluencies<br>*Surprisal | .024               | 23   | 4.28                         | −54 | −22 | 6  | LSTG                      |
| Silent disfluencies<br>*Frequency | 0                  | 190  | 5.85                         | −58 | −16 | 2  | LSTG                      |
|                                   | 0                  | 56   | 5.24                         | −52 | 20  | 22 | LIFG pars triangularis    |
|                                   | .013               | 25   | 3.97                         | −54 | −42 | 26 | LSMG                      |
| Filled disfluencies               | 0                  | 85   | 7.5                          | 48  | −22 | 10 | RSTG                      |
|                                   | 0                  | 104  | 6.73                         | −52 | −22 | 14 | LSTG                      |
|                                   | .02                | 23   | −5.87                        | −60 | −54 | 26 | LSMG                      |
|                                   | 0                  | 58   | −4.94                        | 56  | −42 | 22 | RSTG                      |
| Filled disfluencies<br>*Surprisal | .038 (uncorrected) | 7    | −5.29                        | 8   | −12 | −2 | R Thalamus                |
| *Surprisal                        | .012 (uncorrected) | 11   | −4.54                        | −60 | −48 | 18 | LSTG                      |
|                                   | .012 (uncorrected) | 11   | −4.07                        | −46 | −54 | 30 | LAG                       |
|                                   | .016 (uncorrected) | 10   | −3.77                        | −54 | −30 | −6 | LMTG                      |
| Filled disfluencies<br>*Frequency | .001               | 37   | 4.89                         | −58 | −40 | 30 | LSMG                      |
|                                   | .009               | 27   | 3.81                         | −36 | −30 | 18 | LSTG                      |

the listener to hear less expected content, as listeners are known to adapt to the distributional aspect of the incoming linguistic input (Arnold et al., 2004, 2007; Bosker et al., 2014, 2019; Corley et al., 2007). The increase in attention may be bottom-up due to the saliency of the disfluency and interruption to the flow of speech. The increased attentional resources dedicated to the processing of incoming input could in turn reduce word processing effort for incoming speech (Collard et al., 2008; Fraundorf & Watson, 2011). Both explanations for the increase in attentional resources are plausible since disfluencies themselves elicited stronger responses to the following word, which could be due to increased bottom-up processing as well as top-down modulation.

The cluster for surprisal extended more broadly in the superior temporal cortex, similarly to previous studies, including to areas beyond auditory cortex that are typically associated with higher-level linguistic processing (Bhattachali & Resnik, 2021; Lopopolo et al., 2017; Matchin & Hickok, 2020; Russo et al., 2020; Willems et al., 2016). The interaction between surprisal and disfluency was found in a cluster in the left posterior middle temporal gyrus and angular gyrus, which has been associated with semantic processing (Matchin, Liao, Gaston, & Lau, 2019; Pallier, Devauchelle, & Dehaene, 2011; Xu et al., 2018). The reduced surprisal effect following disfluencies suggests that listeners responded to surprisal less strongly than predicted if there had been no disfluencies. Note that the decreased effect of surprisal following disfluencies could both mean that the response to high surprisal words is reduced, and/or that the response to low surprisal words increases following disfluencies. Both are plausible, since it may also be more surprising to hear highly predictable words following disfluencies. A decreased response to high surprisal words following disfluencies may index decreased efforts or increased success in integrating less predictable content in discourse (Hagoort, 2013, 2019), as also shown previously by

an N400 reduction to unpredictable words following disfluencies (Corley et al., 2007). Interestingly, the reduction to the effect of surprisal did not overlap with the cluster showing a response to surprisal. This suggests that it is not just the effort associated with processing less predictable words that is affected by disfluencies (i.e. lexical access), but potentially their semantic integration in context (Huizeling et al., 2026). The same area was previously found to be involved in the processing of event and thematic role structure (Giglio, Hagoort, & Ostarek, 2024; Matchin et al., 2019; Pallier et al., 2011).

It should be noted that word surprisal was estimated from a model that was trained on text and as such did not include any information on disfluencies. Therefore, it is possible that the estimates of surprisal do not reflect well what is experienced by listeners, since they ignore all the information that is available in the speech, such as prosody, speech rate, pauses, as well as all the discourse markers and fillers that are usually not present in text. Our results, however, clearly show that the surprisal estimates from text are good enough to predict brain activation when listening to spontaneous speech, given the strong modulation of brain activity in the superior temporal cortex due to surprisal. Therefore, the negative response to surprisal following disfluencies shows that listeners were less surprised than predicted by a model that does not take disfluencies into account.

The response to word frequency resembles previous studies (Huizeling, Arana, et al., 2022; Shain et al., 2020; Slaats et al., 2023). Lower word frequency elicited stronger activity in the mid superior temporal gyrus and in a cluster in middle frontal areas. This response was stronger after disfluencies, meaning that there was an increased effect of frequency following disfluencies. Note that given the interaction with a continuous variable we are not able to distinguish what drove the interaction (i.e. a stronger increase/decrease for low/high

frequency words in either condition), but we can only conclude that the difference in response to low and high frequency words increased following disfluencies. Previous EEG studies found that the effect of word frequency on the N400 is reduced with increased context and predictability (Alday, Schlesewsky, & Bornkessel-Schlesewsky, 2017; Dambacher, Kliegl, Hofmann, & Jacobs, 2006; Payne, Lee, & Federmeier, 2015; Sereno, Hand, Shahid, Mackenzie, & Leuthold, 2020; Van Petten & Kutas, 1990). Similarly, previous MEG studies found that the effect of frequency interacts with sentence context as well as surprisal (Huizeling, Arana, et al., 2022; Slaats et al., 2023). Here, the disfluency marks that the upcoming word is less predictable. The effect of context is thus reduced, which may drive the increase in the effect of frequency. Overall, the current results point to disfluencies preparing the listener for upcoming lexical difficulties in the input, which results in more processing resources allocated to lexical access following disfluencies, which in turn may result in easier integration in context.

The increased effect of frequency following disfluencies could have also been driven by a stronger reduction in the response to high frequency words following disfluencies. This explanation is supported by the frequency and length characteristics of words following disfluencies (Fig. 1), as words following silent pauses were slightly more frequent and shorter than fluent words. This unexpected pattern for the occurrence of disfluencies may be explained by the fact that disfluencies were likely to occur at phrase boundaries (see Fig. 2), respecting intonation units (Clark & Fox Tree, 2002; Fraundorf & Watson, 2014) and as such may be often followed by determiners or discourse markers that are very frequent (e.g. “so”, “then”). The disfluency may thus index upcoming complexity in the upcoming discourse, such as the noun following a determiner in a noun phrase, rather than strictly in the following word. Therefore, it is possible that by looking at the effect of disfluencies on the next word only, we missed some of the effects of disfluencies on language comprehension. Disfluencies may index planning demands that span the message- and grammatical-level (Fraundorf & Watson, 2014) and as such affect the processing of the whole sentence rather than just lexical retrieval for a single word. Future studies may investigate whether the effect of disfluencies on brain activity spans several words or is limited to upcoming lexical items, and whether it also facilitates the processing of complex syntactic constructions.

In addition, we asked whether the effects of silent and filled disfluencies on upcoming linguistic processing differ, as their occurrence in speech can differ (Clark & Fox Tree, 2002; Fraundorf & Watson, 2014). Silent pauses often occur at syntactic boundaries, and with difficulty in grammatical and phonological planning (Hawkins, 1971). Fillers instead occur with conceptual difficulties in planning (Fraundorf & Watson, 2014). Some studies also suggest that *uh* and *uhm* have different functions in speech and have different effects on listeners (Clark & Fox Tree, 2002; Fox Tree, 2001). In the current study, entropy and surprisal were highest after filled disfluencies, followed by silent disfluencies, with words not preceded by disfluencies being lower in surprisal and entropy on average. The whole-brain results show that silent and filled disfluencies affected the response to the following word in a

similar way, by increasing the response of the superior temporal gyrus. Silent disfluencies additionally led to increased activation in the left inferior frontal cortex, which could suggest they have a role in syntactic processing (Giglio, Ostarek, et al., 2024; Hagoort, 2013). Silent and filled disfluencies also seem to modulate the word frequency effect in a similar way, although only the effect of silent and not filled disfluencies extended along the temporal lobe and to the left inferior frontal cortex. The negative interaction between surprisal and disfluencies was only found with filled disfluencies (with  $p < .05$  uncorrected for multiple comparisons). Silent disfluencies instead increased the effect of surprisal in the left superior temporal gyrus. Since the combination of the different types of disfluencies led to a stronger negative cluster, it is likely that silent and filled disfluencies shared some function, which was not strong enough to appear when considering them separately. The dissociation in their effects suggests that silent disfluencies enhanced word processing, while filled disfluencies may have facilitated word integration in context, since they led to decreased activation in relation to word surprisal in a brain region often associated with event and thematic role structure processing (Giglio, Hagoort, & Ostarek, 2024; Matchin et al., 2019). It is also possible that silent disfluencies may have had different effects when they occurred at sentence boundaries (which may have been related to parsing or message-level difficulties) relative to when they occurred within phrases, which may have been more specifically related to lexical complexity. Future studies using higher temporal resolution methods and longer speech samples may be able to better distinguish the effects of filled and silent hesitations in discourse comprehension, by for example separating disfluencies also based on where they occur in speech, which is likely related to different difficulties in planning.

The rate of disfluencies found here was somewhat higher than reported in previous studies (around 6% of words, e.g. Bortfeld et al., 2001; Fox Tree, 1995). Previous studies that reported the rate of disfluencies usually focused on filled disfluencies only. In the current study, 11% of words were preceded by a filled disfluency, while 15% of words were preceded by a silent disfluency. It is possible that the rate of 11% is higher than previous studies for a few reasons. The speech was elicited during fMRI scanning, which is an uncomfortable environment in which to speak, as participants are asked to speak lying down and during noisy fMRI acquisition. In addition, a study by Bortfeld et al. (2001) reported that disfluencies tend to increase with longer turns. Since these were monologues, it is unsurprising that the disfluency rate was higher (and tended to increase throughout the production, Fig. 1A). Finally, the speaker was asked to recall the events of a television series, and it is likely that they were hesitating while trying to remember events, as opposed to speaking contexts usually investigated in previous studies (e.g. referential communication tasks or naturalistic corpora).

In conclusion, in an fMRI study of narrative comprehension, we investigated the effects of disfluencies on the neural activity associated with word frequency and surprisal. This study took advantage of spontaneously produced speech, which included disfluencies, as a naturalistic stimulus, to investigate the role of disfluencies in naturalistic language

processing. Thanks to the use of fMRI as a readout of brain activity, we could focus on incremental processing and additionally uncover networks involved in the comprehension of spontaneous speech. Words preceded by disfluencies were associated with stronger activity in the superior temporal gyrus relative to words spoken fluently. The effect of word frequency was stronger following disfluencies, while the response to word surprisal decreased following disfluencies. The modulations in the effects of frequency and surprisal suggest that disfluencies in spontaneous speech are informative for the listener, extending and confirming previous evidence that was primarily based on artificial and experimentally controlled stimuli (Collard et al., 2008; Fraundorf & Watson, 2011; Huizeling, Peeters, & Hagoort, 2022).

### CRediT authorship contribution statement

**Laura Giglio:** Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Peter Hagoort:** Writing – review & editing, Resources, Conceptualization. **Eleanor Huizeling:** Writing – review & editing, Visualization, Methodology, Data curation, Conceptualization.

### Data and code availability

The data used in this study is openly available on OpenNeuro (<https://openneuro.org/datasets/ds001110/versions/000003>). The linguistic annotations and code used in this study are available on OSF repository: [https://osf.io/zqe2f/overview?view\\_only=780bda0d3ce04817a0ba504c9a1e8727](https://osf.io/zqe2f/overview?view_only=780bda0d3ce04817a0ba504c9a1e8727).

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### Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cortex.2026.02.017>.

### REFERENCES

- Agmon, G., Cho, S., Ash, S., Cousins, K. A. Q., Blennow, K., Zetterberg, H., et al. (2025). Automatic quantification of syntactic complexity in natural spontaneous speech of people with primary progressive aphasia. *Aphasiology*, 1–22. <https://doi.org/10.1080/02687038.2025.2462282>
- Agmon, G., Jaeger, M., Magen, E., Pinto, D., Perelmuter, Y., Zion Golumbic, E., et al. (2025). Challenges and methods in annotating natural speech for neurolinguistic research. *Neurobiology of Language*, 1–27. <https://doi.org/10.1162/nol.a.12>
- Agmon, G., Jaeger, M., Tsarfaty, R., Bleichner, M. G., & Zion Golumbic, E. (2023). “Um..., it's really difficult to... um... speak fluently”: Neural tracking of spontaneous speech. *Neurobiology of Language*, 4(3), 435–454. [https://doi.org/10.1162/nol.a\\_00109](https://doi.org/10.1162/nol.a_00109)
- Agmon, G., Pradhan, S., Ash, S., Nevler, N., Liberman, M., Grossman, M., et al. (2024). Automated measures of syntactic complexity in natural speech production: Older and younger adults as a case study. *Journal of Speech, Language, and Hearing Research: JSLHR*, 67(2), 545–561. <https://doi.org/10.1044/2023.JSLHR-23-00009>
- Alday, P. M., Schlesewsky, M., & Bornkessel-Schlesewsky, I. (2017). Electrophysiology reveals the neural dynamics of naturalistic auditory language processing: Event-related potentials reflect continuous model updates. *eNeuro*, 4(6). <https://doi.org/10.1523/ENEURO.0311-16.2017>
- Andersen, G. (1998). The pragmatic marker like from a relevance-theoretic perspective. *Discourse Markers: Descriptions and Theory*, 57, 147.
- Armeni, K., Willems, R. M., van den Bosch, A., & Schoffelen, J.-M. (2019). Frequency-specific brain dynamics related to prediction during language comprehension. *Neuroimage*, 198, 283–295. <https://doi.org/10.1016/j.neuroimage.2019.04.083>
- Arnold, J. E., Kam, C. L. H., & Tanenhaus, M. K. (2007). If you say thee uh you are describing something hard: The on-line attribution of disfluency during reference comprehension. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 33(5), 914–930. <https://doi.org/10.1037/0278-7393.33.5.914>
- Arnold, J. E., Tanenhaus, M. K., Altmann, R. J., & Fagnano, M. (2004). The old and thee, uh, new: Disfluency and reference resolution. *Psychological Science*, 15(9), 578–582. <https://doi.org/10.1111/j.0956-7976.2004.00723.x>
- Avants, B. B., Epstein, C. L., Grossman, M., & Gee, J. C. (2008). Symmetric diffeomorphic image registration with cross-correlation: Evaluating automated labeling of elderly and neurodegenerative brain. *Medical Image Analysis*, 12(1), 26–41. <https://doi.org/10.1016/j.media.2007.06.004>
- Bailey, K. G. D., & Ferreira, F. (2003). Disfluencies affect the parsing of garden-path sentences. *Journal of Memory and Language*, 49(2), 183–200. [https://doi.org/10.1016/S0749-596X\(03\)00027-5](https://doi.org/10.1016/S0749-596X(03)00027-5)
- Barr, D. J., & Seyfeddinipur, M. (2010). The role of fillers in listener attributions for speaker disfluency. *Language and Cognitive Processes*. <https://doi.org/10.1080/01690960903047122> (world).
- Bates, D., Mächler, M., Bolker, B., & Walker, S. (2015). Fitting linear mixed-effects models using lme4. *Journal of Statistical Software*, 67(1), Article 1. <https://doi.org/10.18637/jss.v067.i01>
- Beattie, G. W., & Butterworth, B. L. (1979). Contextual probability and word frequency as determinants of pauses and errors in spontaneous speech. *Language and Speech*, 22(3), 201–211. <https://doi.org/10.1177/002383097902200301>
- Behzadi, Y., Restom, K., Liao, J., & Liu, T. T. (2007). A component based noise correction method (CompCor) for BOLD and perfusion based fMRI. *Neuroimage*, 37(1), 90–101. <https://doi.org/10.1016/j.neuroimage.2007.04.042>
- Bhattachali, S., Fabre, M., Luh, W.-M., Saied, H. A., Constant, M., Pallier, C., et al. (2019). Localising memory retrieval and syntactic composition: An fMRI study of naturalistic language comprehension. *Language, Cognition and Neuroscience*, 34(4), 491–510. <https://doi.org/10.1080/23273798.2018.1518533>
- Bhattachali, S., & Resnik, P. (2021). Using surprisal and fMRI to map the neural bases of broad and local contextual prediction during natural language comprehension. *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*, 3786–3798. <https://doi.org/10.18653/v1/2021.findings-acl.332>
- Born, S. J., Shi, K., Lee Masson, H., Lee, H., Lee, Y., & Chen, J. (2023). Word-timestamped transcripts of two spoken narrative recall

- functional neuroimaging datasets. *Data in Brief*, 50, Article 109490. <https://doi.org/10.1016/j.dib.2023.109490>
- Bortfeld, H., Leon, S. D., Bloom, J. E., Schober, M. F., & Brennan, S. E. (2001). Disfluency rates in conversation: Effects of age, relationship, topic, role, and gender. *Language and Speech*, 44(2), 123–147. <https://doi.org/10.1177/00238309010440020101>
- Bosker, H. R., Quené, H., Sanders, T., & de Jong, N. H. (2014). Native 'um's elicit prediction of low-frequency referents, but non-native 'um's do not. *Journal of Memory and Language*, 75, 104–116. <https://doi.org/10.1016/j.jml.2014.05.004>
- Bosker, H. R., van Os, M., Does, R., & van Bergen, G. (2019). Counting 'uhm's: How tracking the distribution of native and non-native disfluencies influences online language comprehension. *Journal of Memory and Language*, 106, 189–202. <https://doi.org/10.1016/j.jml.2019.02.006>
- Brennan, J. R. (2016). Naturalistic sentence comprehension in the brain. *Language and Linguistics Compass*, 10(7), 299–313. <https://doi.org/10.1111/lnc3.12198>
- Brennan, J. R., Stabler, E. P., Van Wagenen, S. E., Luh, W.-M., & Hale, J. T. (2016). Abstract linguistic structure correlates with temporal activity during naturalistic comprehension. *Brain and Language*, 157–158, 81–94. <https://doi.org/10.1016/j.bandl.2016.04.008>
- Brysaert, M., New, B., & Keuleers, E. (2012). Adding part-of-speech information to the SUBTLEX-US word frequencies. *Behavior Research Methods*, 44(4), 991–997. <https://doi.org/10.3758/s13428-012-0190-4>
- Chen, J., Leong, Y. C., Honey, C. J., Yong, C. H., Norman, K. A., & Hasson, U. (2017). Shared memories reveal shared structure in neural activity across individuals. *Nature Neuroscience*, 20(1), Article 1. <https://doi.org/10.1038/nn.4450>
- Chen, J., Leong, Y. C., Honey, C. J., Yong, C. H., Norman, K. A., & Hasson, U. (2018). Sherlock. OpenNeuro [Dataset] <https://openneuro.org/datasets/ds001132/versions/00003>
- Clark, H. H., & Fox Tree, J. E. (2002). Using uh and um in spontaneous speaking. *Cognition*, 84(1), 73–111. [https://doi.org/10.1016/S0010-0277\(02\)00017-3](https://doi.org/10.1016/S0010-0277(02)00017-3)
- Collard, P., Corley, M., MacGregor, L. J., & Donaldson, D. I. (2008). Attention orienting effects of hesitations in speech: Evidence from ERPs. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 34(3), 696–702. <https://doi.org/10.1037/0278-7393.34.3.696>
- Coopmans, C. W., Hoop, H. de, Tezcan, F., Hagoort, P., & Martin, A. E. (2025). Language-specific neural dynamics extend syntax into the time domain. *PLOS Biology*, 23(1), Article e3002968. <https://doi.org/10.1371/journal.pbio.3002968>
- Corley, M., & Hartsuiker, R. J. (2011). Why Um Helps Auditory Word Recognition: The Temporal Delay Hypothesis. *PLOS ONE*, 6(5), Article e19792. <https://doi.org/10.1371/journal.pone.0019792>
- Corley, M., MacGregor, L. J., & Donaldson, D. I. (2007). It's the way that you, er, say it: Hesitations in speech affect language comprehension. *Cognition*, 105(3), 658–668. <https://doi.org/10.1016/j.cognition.2006.10.010>
- Dale, A. M., Fischl, B., & Sereno, M. I. (1999). Cortical surface-based analysis: I. Segmentation and surface reconstruction. *Neuroimage*, 9(2), 179–194. <https://doi.org/10.1006/nimg.1998.0395>
- Dambacher, M., Kliegl, R., Hofmann, M., & Jacobs, A. M. (2006). Frequency and predictability effects on event-related potentials during reading. *Brain Research*, 1084(1), 89–103. <https://doi.org/10.1016/j.brainres.2006.02.010>
- Dammalapati, S., Rajkumar, R., Ranjan, S., & Agarwal, S. (2021). Effects of duration, locality, and surprisal in speech disfluency prediction in English spontaneous speech. In A. Ettinger, E. Pavlick, & B. Prickett (Eds.), *Proceedings of the Society for computation in linguistics 2021* (pp. 91–101). Association for Computational Linguistics. <https://aclanthology.org/2021.scil-1.9/>
- Diachek, E., & Brown-Schmidt, S. (2023). The effect of disfluency on memory for what was said. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 49(8), 1306–1324. <https://doi.org/10.1037/xlm0001156>
- Diskin, C. (2017). The use of the discourse-pragmatic marker 'like' by native and non-native speakers of English in Ireland. *Journal of Pragmatics*, 120, 144–157. <https://doi.org/10.1016/j.pragma.2017.08.004>
- Eklund, R., & Ingvar, M. (2016). Supplementary motor area activation in disfluency perception: An fMRI study of listener neural responses to spontaneously produced unfilled and filled pauses (pp. 1378–1381). <https://urn.kb.se/resolve?urn=urn:nbn:se:liu:diva-135032>
- Esteban, O., Markiewicz, C. J., Blair, R. W., Moodie, C. A., Isik, A. I., Erramuzpe, A., et al. (2018). fMRIPrep: A robust preprocessing pipeline for functional MRI. *Nature Methods*, 1. <https://doi.org/10.1038/s41592-018-0235-4>
- Fox Tree, J. E. (1995). The effects of false starts and repetitions on the processing of subsequent words in spontaneous speech. *Journal of Memory and Language*, 34(6), 709–738. <https://doi.org/10.1006/jmla.1995.1032>
- Fox Tree, J. E. (2001). Listeners' uses of um and uh in speech comprehension. *Memory & Cognition*, 29(2), 320–326. <https://doi.org/10.3758/BF03194926>
- Fox Tree, J. E. (2007). Folk notions of um and uh, you know, and like. *Tüberküloz Ve Toraks*, 27(3), 297–314. <https://doi.org/10.1515/TEXT.2007.012> (25882331).
- Fox Tree, J. E. (2010). Discourse markers across speakers and settings. *Language and Linguistics Compass*, 4(5), 269–281. <https://doi.org/10.1111/j.1749-818X.2010.00195.x>
- Fraundorf, S. H., & Watson, D. G. (2011). The disfluent discourse: Effects of filled pauses on recall. *Journal of Memory and Language*, 65(2), 161–175. <https://doi.org/10.1016/j.jml.2011.03.004>
- Fraundorf, S. H., & Watson, D. G. (2014). Alice's adventures in um-derland: Psycholinguistic sources of variation in disfluency production. *Language, Cognition and Neuroscience*, 29(9), 1083–1096. <https://doi.org/10.1080/01690965.2013.832785>
- Giglio, L., Hagoort, P., & Ostarek, M. (2024). Neural encoding of semantic structures during sentence production. *Cerebral Cortex*, 34(12), bhae482. <https://doi.org/10.1093/cercor/bhae482>
- Giglio, L., Ostarek, M., Sharoh, D., & Hagoort, P. (2024). Diverging neural dynamics for syntactic structure building in naturalistic speaking and listening. *Proceedings of the National Academy of Sciences*, 121(11), Article e2310766121. <https://doi.org/10.1073/pnas.2310766121>
- Goldman-Eisler, F. (1958a). Speech analysis and mental processes. *Language and Speech*, 1(1), 59–75. <https://doi.org/10.1177/002383095800100105>
- Goldman-Eisler, F. (1958b). The predictability of words in context and the length of pauses in speech. *Language and Speech*, 1(3), 226–231. <https://doi.org/10.1177/002383095800100308>
- Goldman-Eisler, F. (1968). *Psycholinguistics: Experiments in spontaneous speech*. Academic Press.
- Gorgolewski, K. J., Burns, C. D., Madison, C., Clark, D., Halchenko, Y. O., Waskom, M. L., et al. (2011). Nipype: A flexible, lightweight and extensible neuroimaging data processing framework in Python. *Frontiers in Neuroinformatics*, 5, 13. <https://doi.org/10.3389/fninf.2011.00013>
- Gorgolewski, K. J., Esteban, O., Markiewicz, C. J., Ziegler, E., Ellis, D. G., Notter, M. P., et al. (2018). Nipype. Software. <https://doi.org/10.5281/zenodo.596855>
- Greve, D. N., & Fischl, B. (2009). Accurate and robust brain image alignment using boundary-based registration. *Neuroimage*, 48(1), 63–72. <https://doi.org/10.1016/j.neuroimage.2009.06.060>

- Hagoort, P. (2013). MUC (memory, unification, control) and beyond. *Frontiers in Psychology*, 4, 416. <https://doi.org/10.3389/fpsyg.2013.00416>
- Hagoort, P. (2019). The neurobiology of language beyond single-word processing. *Science*, 366(6461), 55–58. <https://doi.org/10.1126/science.aax0289>
- Hasson, U., Egidi, G., Marelli, M., & Willems, R. M. (2018). Grounding the neurobiology of language in first principles: The necessity of non-language-centric explanations for language comprehension. *Cognition*, 180, 135–157. <https://doi.org/10.1016/j.cognition.2018.06.018>
- Hausfeld, L., Hamers, I. M. H., & Formisano, E. (2024). fMRI speech tracking in primary and non-primary auditory cortex while listening to noisy scenes. *Communications Biology*, 7(1), 1217. <https://doi.org/10.1038/s42003-024-06913-z>
- Hawkins, P. R. (1971). The syntactic location of hesitation pauses. *Language and Speech*, 14(3), 277–288. <https://doi.org/10.1177/002383097101400308>
- Heilbron, M., Armeni, K., Schoffelen, J.-M., Hagoort, P., & de Lange, F. P. (2022). A hierarchy of linguistic predictions during natural language comprehension. *Proceedings of the National Academy of Sciences*, 119(32), Article e2201968119. <https://doi.org/10.1073/pnas.2201968119>
- Heyselaar, E., Peeters, D., & Hagoort, P. (2021). Do we predict upcoming speech content in naturalistic environments? *Language, Cognition and Neuroscience*, 36(4), 440–461. <https://doi.org/10.1080/23273798.2020.1859568>
- Holmes, J. (1990). Hedges and boosters in women's and men's speech. *Language & Communication*, 10(3), 185–205. [https://doi.org/10.1016/0271-5309\(90\)90002-S](https://doi.org/10.1016/0271-5309(90)90002-S)
- Huizeling, E., Alday, P. M., Peeters, D., & Hagoort, P. (2023). Combining EEG and 3D-eye-tracking to study the prediction of upcoming speech in naturalistic virtual environments: A proof of principle. *Neuropsychologia*, 191, Article 108730. <https://doi.org/10.1016/j.neuropsychologia.2023.108730>
- Huizeling, E., Arana, S., Hagoort, P., & Schoffelen, J.-M. (2022). Lexical frequency and sentence context influence the brain's response to single words. *Neurobiology of Language*, 3(1), 149–179. [https://doi.org/10.1162/nol\\_a\\_00054](https://doi.org/10.1162/nol_a_00054)
- Huizeling, E., Peeters, D., & Hagoort, P. (2021). Prediction of upcoming speech under fluent and disfluent conditions: Eye tracking evidence from immersive virtual reality. *Language, Cognition and Neuroscience*, 0(0), 1–28. <https://doi.org/10.1080/23273798.2021.1994621>
- Huizeling, E., Peeters, D., & Hagoort, P. (2022). Prediction of upcoming speech under fluent and disfluent conditions: Eye tracking evidence from immersive virtual reality. *Language, Cognition and Neuroscience*, 37(4), 481–508. <https://doi.org/10.1080/23273798.2021.1994621>
- Huizeling, A., Peeters, & Hagoort. (2026). To what extent do pragmatic cues from disfluencies inform our predictions of spoken language during naturalistic language processing? *Psyarxiv*.
- Jaeger, T. F. (2010). Redundancy and reduction: Speakers manage syntactic information density. *Cognitive Psychology*, 61(1), 23–62. <https://doi.org/10.1016/j.cogpsych.2010.02.002>
- Jenkinson, M., Bannister, P., Brady, M., & Smith, S. (2002). Improved optimization for the robust and accurate linear registration and motion correction of brain images. *Neuroimage*, 17(2), 825–841. <https://doi.org/10.1006/nimg.2002.1132>
- Jucker, A. H., Smith, S. W., & Lüdge, T. (2003). Interactive aspects of vagueness in conversation. *Journal of Pragmatics*, 35(12), 1737–1769. [https://doi.org/10.1016/S0378-2166\(02\)00188-1](https://doi.org/10.1016/S0378-2166(02)00188-1)
- Klein, A., Ghosh, S. S., Bao, F. S., Giard, J., Häme, Y., Stavsky, E., et al. (2017). Mindboggling morphometry of human brains. *Plos Computational Biology*, 13(2), Article e1005350. <https://doi.org/10.1371/journal.pcbi.1005350>
- Kuznetsova, A., Brockhoff, P. B., Christensen, R. H. B., & Jensen, S. P. (2020). *lmerTest: Tests in Linear mixed effects models* [Computer software] Version 3.1-3. <https://cran.r-project.org/web/packages/lmerTest/index.html>
- Lenth, R. V., Buerkner, P., Herve, M., Love, J., Miguez, F., Riebl, H., et al. (2022). *Emmeans: Estimated marginal means, Aka least-squares means* [Computer software] Version 1.7.3. <https://CRAN.R-project.org/package=emmeans>
- Lopopolo, A., Frank, S. L., Bosch, A. van den, & Willems, R. M. (2017). Using stochastic language models (SLM) to map lexical, syntactic, and phonological information processing in the brain. *PLOS ONE*, 12(5), Article e0177794. <https://doi.org/10.1371/journal.pone.0177794>
- MacGregor, L. J., Corley, M., & Donaldson, D. I. (2010). Listening to the sound of silence: Disfluent silent pauses in speech have consequences for listeners. *Neuropsychologia*, 48(14), 3982–3992. <https://doi.org/10.1016/j.neuropsychologia.2010.09.024>
- Matchin, W., & Hickok, G. (2020). The cortical organization of syntax. *Cerebral Cortex*, 30(3), 1481–1498. <https://doi.org/10.1093/cercor/bhz180>
- Matchin, W., Liao, C.-H., Gaston, P., & Lau, E. (2019). Same words, different structures: An fMRI investigation of argument relations and the angular gyrus. *Neuropsychologia*, 125, 116–128. <https://doi.org/10.1016/j.neuropsychologia.2019.01.019>
- Pallier, C., Devauchelle, A.-D., & Dehaene, S. (2011). Cortical representation of the constituent structure of sentences. *Proceedings of the National Academy of Sciences*, 108(6), 2522–2527. <https://doi.org/10.1073/pnas.1018711108>
- Payne, B. R., Lee, C.-L., & Federmeier, K. D. (2015). Revisiting the incremental effects of context on word processing: Evidence from single-word event-related brain potentials. *Psychophysiology*, 52(11), 1456–1469. <https://doi.org/10.1111/psyp.12515>
- Peeters, D. (2019). Virtual reality: A game-changing method for the language sciences. *Psychonomic Bulletin & Review*, 26(3), 894–900. <https://doi.org/10.3758/s13423-019-01571-3>
- Penny, W. D., Friston, K. J., Ashburner, J. T., Kiebel, S. J., & Nichols, T. E. (2011). *Statistical parametric mapping: The analysis of functional brain images*. Elsevier.
- Power, J. D., Mitra, A., Laumann, T. O., Snyder, A. Z., Schlaggar, B. L., & Petersen, S. E. (2014). Methods to detect, characterize, and remove motion artifact in resting state fMRI. *Neuroimage*, 84(Supplement C), 320–341. <https://doi.org/10.1016/j.neuroimage.2013.08.048>
- Pruim, R. H. R., Mennes, M., van Rooij, D., Llera, A., Buitelaar, J. K., & Beckmann, C. F. (2015). ICA-AROMA: A robust ICA-based strategy for removing motion artifacts from fMRI data. *Neuroimage*, 112(Supplement C), 267–277. <https://doi.org/10.1016/j.neuroimage.2015.02.064>
- Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., & Sutskever, I. (2019). Language models are unsupervised multitask learners. *Open Biology*, 1(8), 9.
- Russo, A. G., De Martino, M., Mancuso, A., Iaconetta, G., Manara, R., Elia, A., et al. (2020). Semantics-weighted lexical surprisal modeling of naturalistic functional MRI time-series during spoken narrative listening. *Neuroimage*, 222, Article 117281. <https://doi.org/10.1016/j.neuroimage.2020.117281>
- Sereno, S. C., Hand, C. J., Shahid, A., Mackenzie, I. G., & Leuthold, H. (2020). Early EEG correlates of word frequency and contextual predictability in reading. *Language, Cognition and Neuroscience*, 35(5), 625–640. <https://doi.org/10.1080/23273798.2019.1580753>
- Shain, C., Blank, I. A., van Schijndel, M., Schuler, W., & Fedorenko, E. (2020). fMRI reveals language-specific predictive coding during naturalistic sentence comprehension.

- Neuropsychologia*, 138, Article 107307. <https://doi.org/10.1016/j.neuropsychologia.2019.107307>
- Slaats, S., Weissbart, H., Schoffelen, J.-M., Meyer, A. S., & Martin, A. E. (2023). Delta-band neural responses to individual words are modulated by sentence processing. *Journal of Neuroscience*, 43(26), 4867–4883. <https://doi.org/10.1523/JNEUROSCI.0964-22.2023>
- Smirnova, K., Korotaev, N., Panikratova, Y., Lebedeva, I., Pechenkova, E., & Fedorova, O. (2020). Using the RUPEx multichannel corpus in a pilot fMRI study on speech disfluencies. In N. Calzolari, F. Béchet, P. Blache, K. Choukri, C. Cieri, T. Declerck, et al. (Eds.), *Proceedings of the twelfth language resources and evaluation conference* (pp. 195–203). European Language Resources Association. <https://aclanthology.org/2020.lrec-1.25>.
- Tromp, J., Peeters, D., Meyer, A. S., & Hagoort, P. (2018). The combined use of virtual reality and EEG to study language processing in naturalistic environments. *Behavior Research Methods*, 50(2), 862–869. <https://doi.org/10.3758/s13428-017-0911-9>
- Tustison, N. J., Avants, B. B., Cook, P. A., Zheng, Y., Egan, A., Yushkevich, P. A., et al. (2010). N4ITK: Improved N3 bias correction. *IEEE Transactions on Medical Imaging*, 29(6), 1310–1320. <https://doi.org/10.1109/TMI.2010.2046908>
- Van Petten, C., & Kutas, M. (1990). Interactions between sentence context and word frequency in event-related brain potentials. *Memory & Cognition*, 18(4), 380–393. <https://doi.org/10.3758/BF03197127>
- Wehbe, L., Murphy, B., Talukdar, P., Fyshe, A., Ramdas, A., & Mitchell, T. (2014). Simultaneously uncovering the patterns of brain regions involved in different story reading subprocesses. *PLOS ONE*, 9(11), Article e112575. <https://doi.org/10.1371/journal.pone.0112575>
- Willems, R. M., Frank, S. L., Nijhof, A. D., Hagoort, P., & van den Bosch, A. (2016). Prediction during natural Language comprehension. *Cerebral Cortex*, 26(6), 2506–2516. <https://doi.org/10.1093/cercor/bhv075>
- Wolf, T., Debut, L., Sanh, V., Chaumond, J., Delangue, C., Moi, A., et al. (2020). Transformers: state-of-the-Art natural language processing. In *Proceedings of the 2020 conference on empirical methods in natural language processing: System demonstrations* (pp. 38–45). <https://doi.org/10.18653/v1/2020.emnlp-demos.6>
- Xu, Y., Wang, X., Wang, X., Men, W., Gao, J.-H., & Bi, Y. (2018). Doctor, teacher, and stethoscope: Neural representation of different types of semantic relations. *The Journal of Neuroscience*, 38(13), 3303–3317. <https://doi.org/10.1523/JNEUROSCI.2562-17.2018>
- Zadbood, A., Chen, J., Leong, Y. C., Norman, K. A., & Hasson, U. (2017). How we transmit memories to other brains: Constructing shared neural representations via communication. *Cerebral Cortex*, 27(10), 4988–5000. <https://doi.org/10.1093/cercor/bhx202>
- Zadbood, A., Chen, J., Leong, Y. C., Norman, K. A., & Hasson, U. (2018). *Sherlock\_Merlin*. *OpenNeuro [Dataset]* <https://openneuro.org/datasets/ds001110/versions/00003>.
- Zhang, Y., Brady, M., & Smith, S. (2001). Segmentation of brain MR images through a hidden markov random field model and the expectation-maximization algorithm. *IEEE Transactions on Medical Imaging*, 20(1), 45–57. <https://doi.org/10.1109/42.906424>